

ANOVA

Analysis of Variance

Introduction

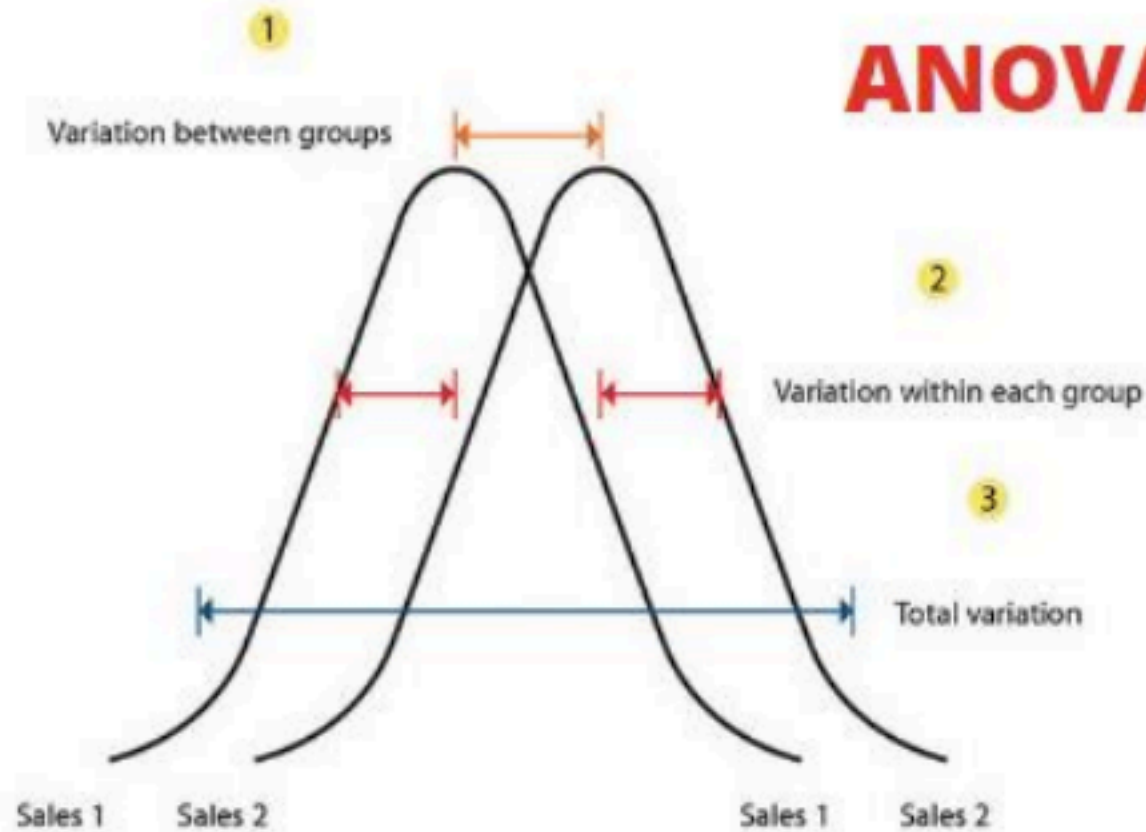
- ANOVA is a type of hypothesis testing which is used to find out the experimental results by analyzing the variance of the different survey groups. It is usually used for deciding the result of the dataset.
- Analysis of variance(ANOVA) is a statistical method to find out if the means of two or more groups are significantly different from each other. It checks the impact of one or more factors by comparing the

means of different samples.

- When we have two samples/groups we use a t-test to find out the mean between those samples but it is not that reliable for more than two samples, therefore, we use ANOVA

ANOVA Testing

ANOVA testing



Sales 1	Sales 2
150	170
155	162
157	177
145	192
130	184
170	169
165	155

Two sample groups
of sales data

Why do we use ANOVA testing?

- In machine learning, the biggest problem is selecting the best features or attributes for training the model. We only require those features that are highly dependent on the response variable so that our model can be able to predict the actual outcome after training the model. ANOVA is used to figure out the result when we have a continuous response variable and the target feature is categorical.

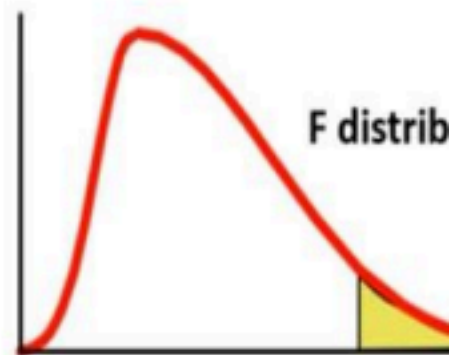
Null Hypothesis – The means for all groups are the same (equal).

$$H_0: \mu_1 = \mu_2 = \mu_3 \dots \mu_n$$

Alternate Hypothesis – The means are different for at **least one pair** of groups

$$H_1: \mu_1 \neq \mu_2 \neq \mu_3 \neq \dots \mu_n$$

$$\underline{\text{ANOVA}} = \frac{\text{Variance Between}}{\text{Variance Within}}$$



Example

- For example, we set up an experience of three groups of people, the very first group gets water drinks, second get some sugary juice and the third one like to take coffee or tea. Now, we need to test everyone's reaction time and want to know if there is any difference between the groups or not.



29
30
31
31
29



28
29
27
30
29



25
28
29
27
29

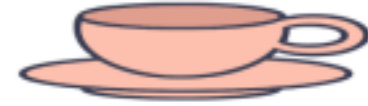
Null Hypothesis



10
12
18
24
36



11
14
19
23
38



12
13
17
25
37

- The **null hypothesis** tells that all the three groups have the same reaction time, we have three groups here to experiment and find out the result so we need to apply the ANOVA testing in case of two groups we could use the t-test when we experiment we would notice that the result won't be same.
- The total variance of all these scores is made up of two parts:

- 1.The variance within the groups: As people have different reaction time in each group.
- 2.The variance between the groups: As the drinks are different which people prefer.

Example one

Example one			
	water	sugersyru	cofee
	10	11	12
	12	14	13
	18	19	17
	24	23	25
	36	38	37
Mean	20	21	20.8
Grand Mean	20.6		

- As we can see here, there is a lot of variation in each sample/group, some of them are faster and some of them

are slower but the groups are quite to one another, there is not much variation between the groups. So we can say that people are making a difference but not the type of drinks, in this case, we need to accept the null hypothesis we can't reject that as the type of drink doesn't put any effect on reaction time.

Example two

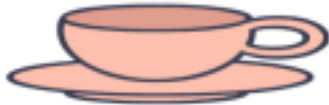
Example two			
	water	sugersyru	cofee
	29	17	10
	29	18	11
	30	19	12
	31	19	13
	31	20	13
Mean	30	18.6	11.8
Grand Mean	20.1333		



29
29
30
31
31



17
18
19
19
20



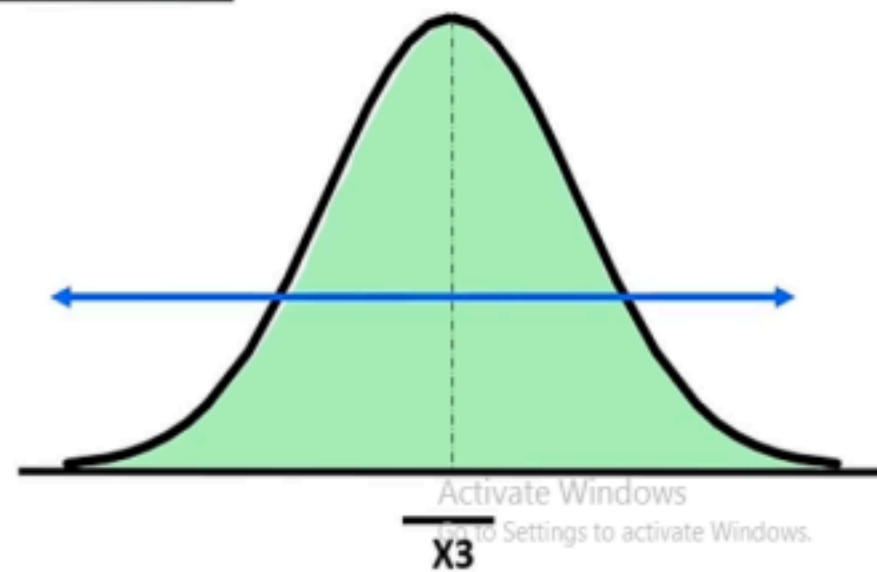
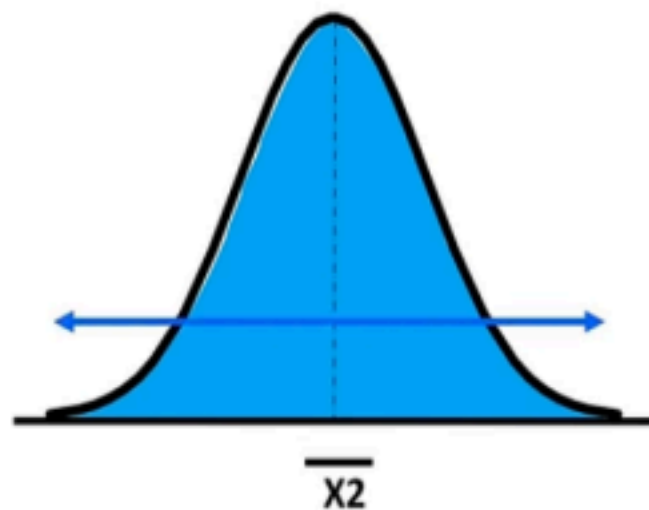
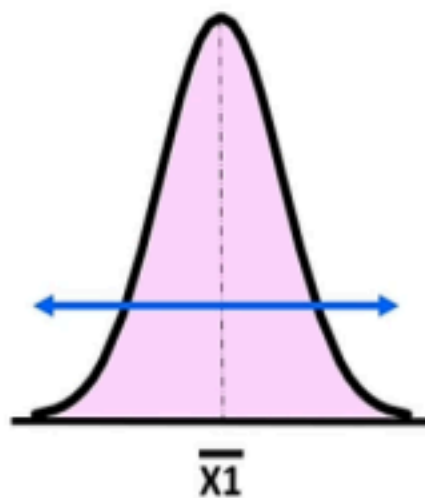
10
11
12
13
13

- Here we can see that there is not much difference within the

groups but there is a lot of difference between the groups. The people's reaction time doesn't make any effect on the groups, so here we will reject the null hypothesis.

Basics of ANOVA

Variability **AROUND/ WITHIN** distribution

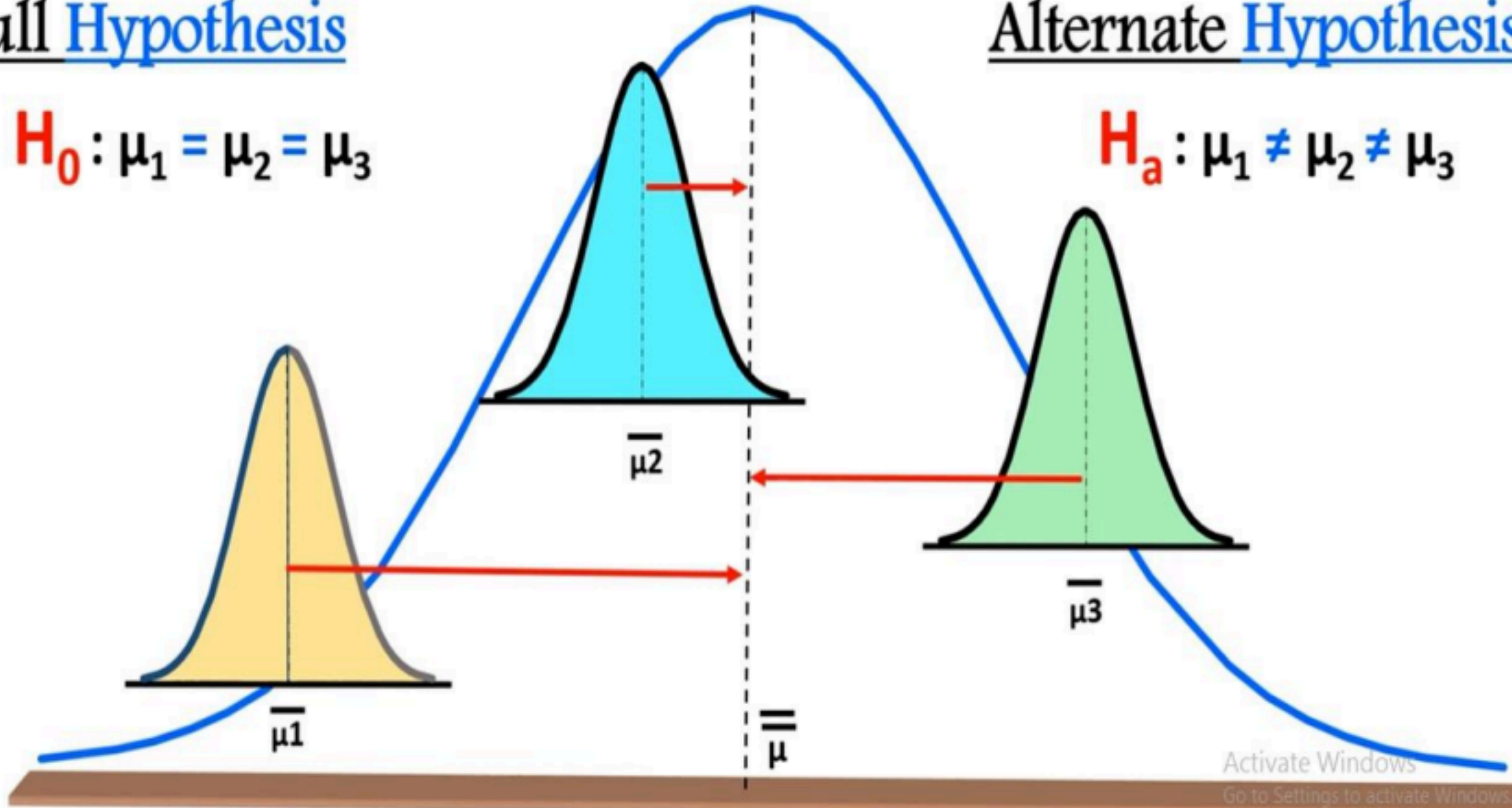


Null Hypothesis

$$H_0: \mu_1 = \mu_2 = \mu_3$$

Alternate Hypothesis

$$H_a: \mu_1 \neq \mu_2 \neq \mu_3$$



Mean, Hypothesis Testing

Mean:

There are two types of mean that we used in ANOVA

- Mean of each sample
- Grand mean is the mean of all the observations combined.

Hypothesis testing:

Hypothesis testing is statistical testing that is used to analyze the assumptions regarding the population parameters. There are two types of hypotheses in the hypothesis testing

- Null hypothesis
- Alternate hypothesis.

Hypothesis in ANOVA is

- $H_0: \mu_1 = \mu_2 = \mu_3 \dots$
- H_1 : Means are not all equal.
- where k = the number of independent comparison groups.

Types of ANOVA

- **One way ANOVA:**

- The one-way ANOVA is used to find out the statistically significant difference between the mean of more than two independent groups.
- More specifically it is used to test the null hypothesis.
- In one-way ANOVA μ = group means and k is a number of groups, if one-way ANOVA returns the significant result, in this case, we accept the alternative hypothesis, this means that the mean of two groups is not equal.

- **Two-way ANOVA:**

- Two-way ANOVA is carried out when you have two independent variables. It is an extension of one-way ANOVA. You can use the two-way ANOVA test when your experiment has a quantitative outcome and there are two independent variables.

- **F-value for ANOVA:**

- The F-value of ANOVA is a tool to help you to determine that, Is the variance between the means of two samples significantly different or not. The ratio of the between the groups and within the groups. It also helps

us to find out the p-Value. The P-value is the probability of getting the result at least at the point where the null hypothesis should be true.

- $F\text{-Value} = \text{Mean between the groups} / \text{Mean within the groups}$

In ANOVA

- In ANOVA, the null hypothesis is that there is no difference among group means. If any group differs significantly from the overall group mean, then the ANOVA will report a statistically significant result.
- Significant differences among group means are calculated using the F statistic, which is the ratio of the mean sum of squares (the variance explained by the independent variable) to the mean square error (the variance left over).

Null Hypothesis – The means for all groups are the same (equal).

$$H_0 : \mu_1 = \mu_2 = \mu_3 \dots \mu_n$$

Alternate Hypothesis – The means are different for at **least one pair** of groups

$$H_1 : \mu_1 \neq \mu_2 \neq \mu_3 \neq \dots \mu_n$$

Means(Grand and Sample)

- As we know, a mean is defined as an arithmetic average of a given range of values. In the ANOVA test, there are two types of mean that are calculated:

- **Grand Mean**:-the average value of sample means of different groups or mean of all the observations combined.
- **Sample Mean**:-A sample mean (μ_n) represents the average value for a group while the grand mean (μ) represents

$$GM (\mu) = \frac{\mu_1 + \mu_2 + \mu_3 + \dots + \mu_n}{n}$$

F-Statistic

- The statistic which measures the extent of difference between the means of different samples or how significantly the means differ is

called the **F-statistic** or **F-Ratio**. It gives us a ratio of the effect we are measuring (in the numerator) and the variation associated with the effect (in the denominator).

$$F = \frac{\text{measure of effect (MSE}_{\text{effect}})}{\text{measure of error (MSE}_{\text{error}})}$$

Sums of Squares

- In statistics, the sum of squares is defined as a statistical technique that is used in regression analysis to determine the dispersion of data

points. In the ANOVA test, it is used while computing the value of F.

- As the sum of squares tells you about the deviation from the mean, it is also known as variation.
- The formula given to calculate the sum of squares is: While calculating the value of F, we need to find SSTotal that is equal to the sum of SS_{Effect} and SS_{Error} .
- $SS_{\text{Total}} = SS_{\text{Effect}} + SS_{\text{Error}}$

Degrees of Freedom (D_f)

- Degrees of Freedom refers to the maximum numbers of logically independent values that have the freedom to vary in a data set.

$$D_f = N - 1$$

where:

D_f = Degrees of Freedom,

N = Number of values

Mean Squared Error (MSE)

- The Mean Squared Error tells us about the average error in a data set. To find the mean squared error, we just divide the sum of squares by the degrees of freedom.

$$MSE = \frac{SS}{D_f}$$

Method-1 (Example)

Sno	Method A	Method B	Method C
1.	10	8	9
2.	9	9	8
3.	8	10	7
4.	7.5	8	10
5.	8.5	8.5	9
6.	9	7	8
7.	10	9.5	7
8.	8	9	10
9.	8	7	9
10.	9	10	8
Group Mean	8.7	8.6	8.5

Overall Mean **8.6**

Between Group Variation = $10 \times (8.7 - 8.6)^2 + 10 \times (8.6 - 8.6)^2 + 10 \times (8.5 - 8.6)^2$

Between Group Variation = 0.2

Within Group Variation: $\sum (X_{ij} - X_j)^2$

Where:

Σ : a symbol that means "sum"
 X_{ij} : the i^{th} observation in group j
 X_j : the mean of group j

Method A: $(10 - 8.7)^2 + (9 - 8.7)^2 + (8 - 8.7)^2 + (7.5 - 8.7)^2 + (8.5 - 8.7)^2 + (9 - 8.7)^2 + (10 - 8.7)^2 + (8 - 8.7)^2 + (8 - 8.7)^2 + (9 - 8.7)^2 = 6.6$

Method B: $(8 - 8.6)^2 + (9 - 8.6)^2 + (10 - 8.6)^2 + (8 - 8.6)^2 + (8.5 - 8.6)^2 + (7 - 8.6)^2 + (9.5 - 8.6)^2 + (9 - 8.6)^2 + (7 - 8.6)^2 + (10 - 8.6)^2 = 10.9$

Method C: $(9 - 8.5)^2 + (8 - 8.5)^2 + (7 - 8.5)^2 + (10 - 8.5)^2 + (9.5 - 8.5)^2 + (8 - 8.5)^2 + (7 - 8.5)^2 + (10 - 8.5)^2 + (9 - 8.5)^2 + (8 - 8.5)^2 = 10.5$

Within Group Variation: $6.6 + 10.9 + 10.5 = 28$

Method-1 (Example) cont...











Real-world application of ANOVA test

- Suppose medical researchers want to find the best diabetes medicine and they have to choose from four medicines. They can choose 20 patients and give them each of the four medicines for four months.
- The researchers can take note of the sugar levels before and after medication for each medicine and then to understand whether there is a statistically significant difference in the mean results from the medications, they can use one-way ANOVA.
- The type of medicine can be a factor and reduction in sugar level can be considered the response. Researchers can then calculate the p value and compare if they are lower than the significance level.

Conclusion

- ANOVA is a statistical hypothesis testing process that tests the significant difference between two or more means. F-value is used to measure the size of the effects while comparing a ratio between those means of the samples/groups.
- The larger f-value shows a high effect, and if the f-value has lower value it shows there is no effect.
- When we noticed there is no effect we reject the null hypothesis and vice versa.

Example in EXCEL























































